

Matrices, Vectors & Complex Numbers



Matrix arithmetic

- 1 Matrix A has rows $(2, 3)$ and $(-1, 4)$; matrix B has rows $(5, -2)$ and $(1, 3)$. Find the rows of $A + B$.
 - 2 Find $3A$ for matrix A with rows $(1, -2)$ and $(4, 0)$.
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Matrix multiplication

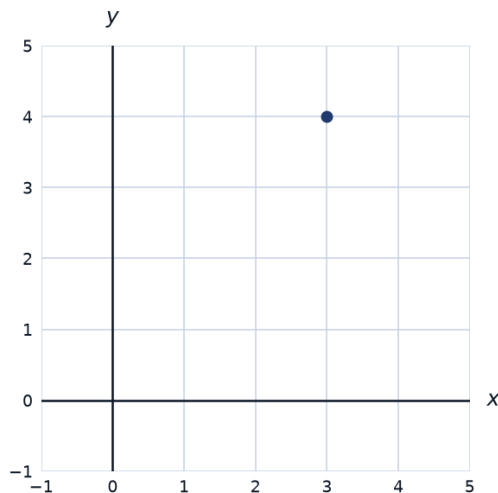
- 3 Matrix A has rows $(1, 2)$ and $(3, 4)$; matrix B has rows $(5, 6)$ and $(7, 8)$. Find AB .
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Determinants and inverses of 2×2 matrices

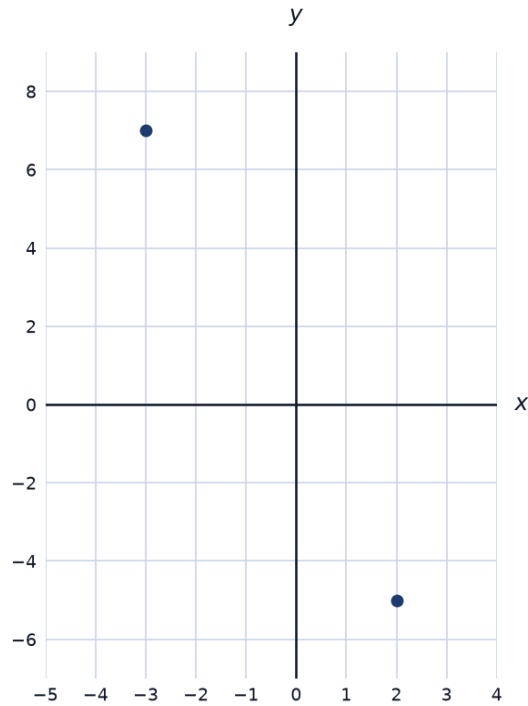
- 4 Find the determinant of the matrix with rows $(4, 2)$ and $(3, 5)$.
 - 5 Find the inverse of the matrix M with rows $(2, 1)$ and $(5, 3)$.
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Vectors: magnitude and direction

- 6 Find the magnitude of the vector $\mathbf{v} = (3, 4)$.

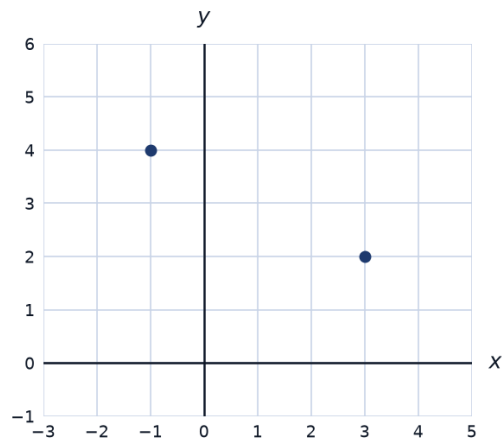


- 7 Given $\mathbf{a} = (2, -5)$ and $\mathbf{b} = (-3, 7)$, find $\mathbf{a} + \mathbf{b}$ and $2\mathbf{a} - \mathbf{b}$.



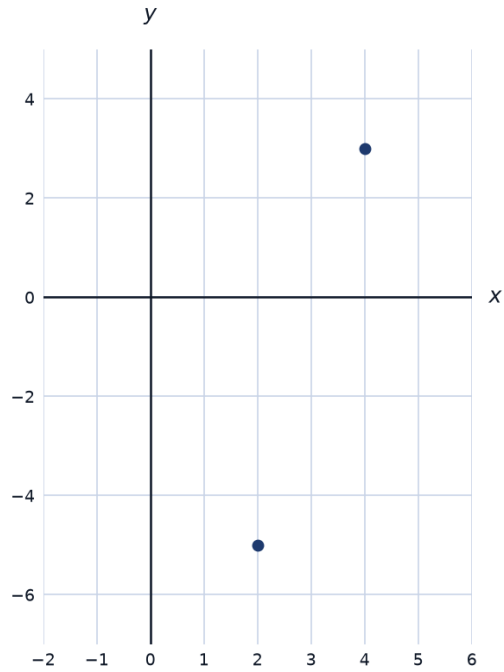
The dot product

- 8 Find $\mathbf{a} \cdot \mathbf{b}$ for $\mathbf{a} = (3, 2)$ and $\mathbf{b} = (-1, 4)$, and determine whether the vectors are perpendicular.

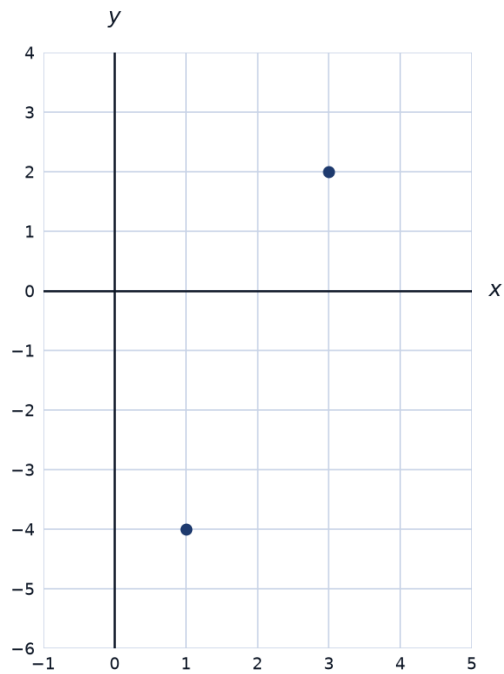


Complex number arithmetic

- 9 Simplify $(4 + 3i) + (2 - 5i)$.

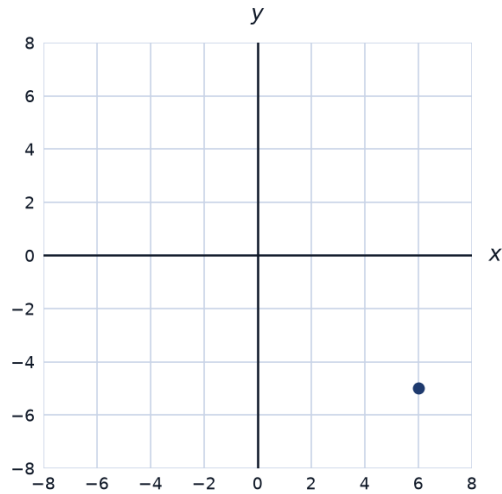


- 10 Simplify $(3 + 2i)(1 - 4i)$.

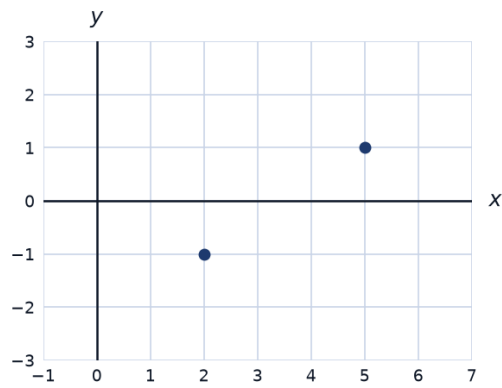


Complex conjugates and division

- 11 Find the complex conjugate of $z = 6 - 5i$, and compute $z\bar{z}$.

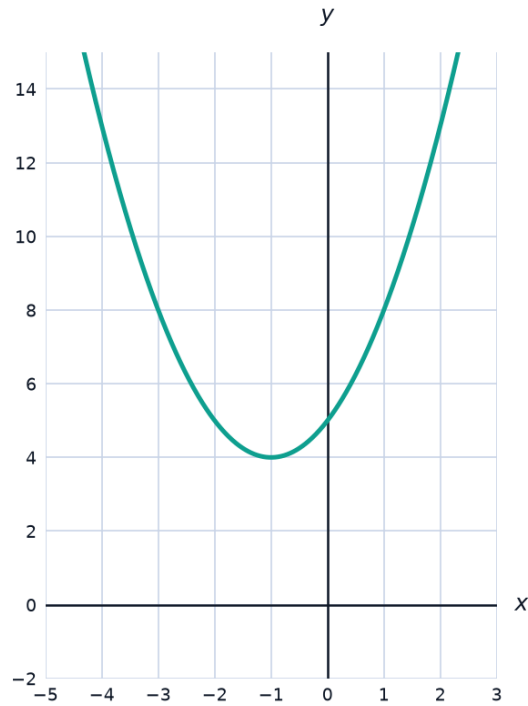


- 12 Simplify $\frac{5+i}{2-i}$ by multiplying by the conjugate of the denominator.



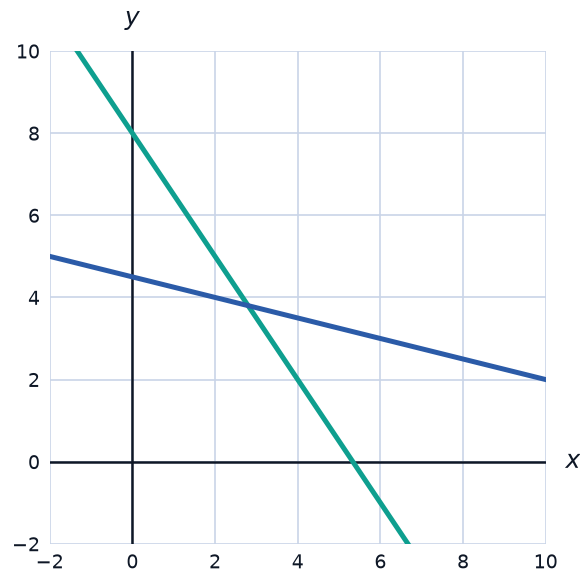
Solving quadratics with complex solutions

- 13 Solve $x^2 + 2x + 5 = 0$, giving your answer in the form $a + bi$.



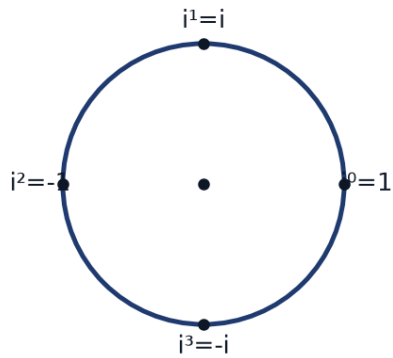
Combining matrices and systems of equations

- 14 This question has two parts. The system $3x + 2y = 16$ and $x + 4y = 18$ can be written as a matrix equation $M\mathbf{v} = \mathbf{c}$, where M has rows $(3, 2)$ and $(1, 4)$. (a) Find $\det(M)$ and M^{-1} . (b) Use M^{-1} to solve the system for x and y .



Powers of i and simplifying complex expressions

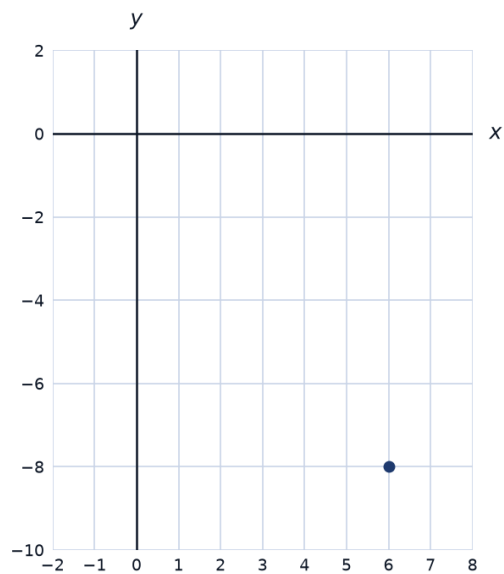
15 Simplify i^{17} , using the fact that powers of i cycle with period 4.



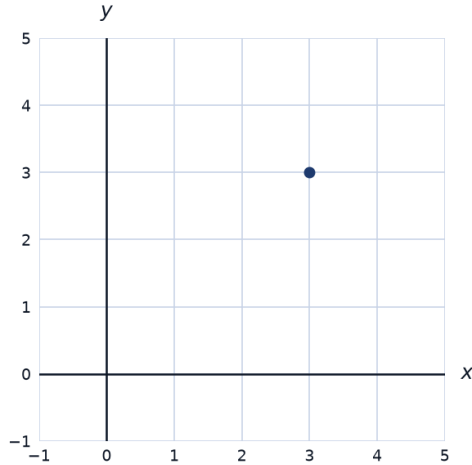
16 Simplify $(2i)^3$.

Vector direction and unit vectors

17 Find a unit vector in the same direction as $\mathbf{v} = (6, -8)$.



- 18 Find the angle a vector $\mathbf{v} = (3, 3)$ makes with the positive x -axis.

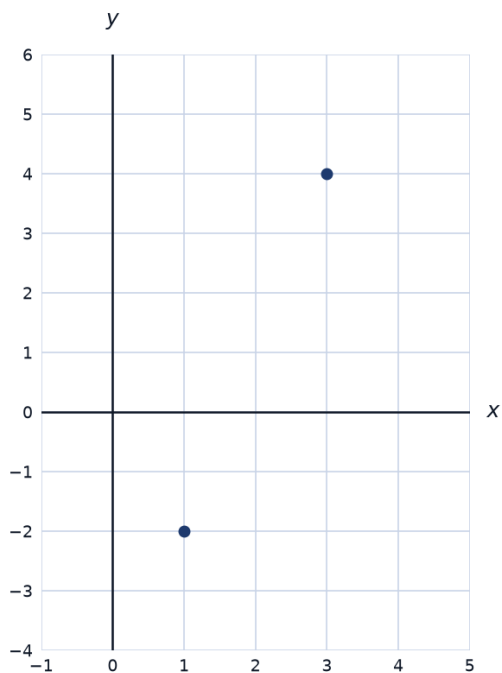


Identity matrix and matrix equations

- 19 Explain why $MI = M$ for any matrix M and identity matrix I of matching size, and verify this for M with rows $(2, 3)$ and $(1, 4)$ multiplied by the 2×2 identity matrix (rows $(1, 0)$ and $(0, 1)$).

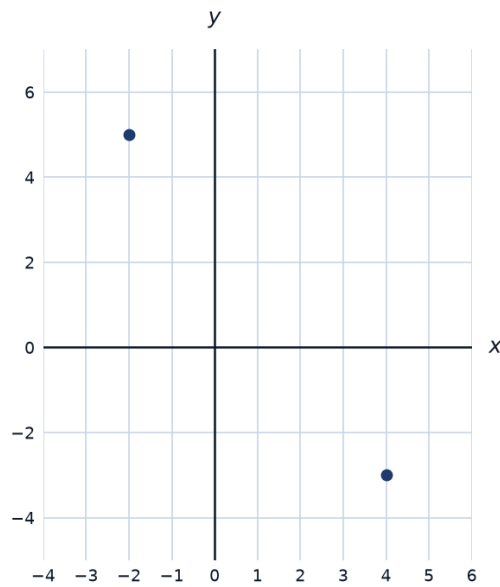
A multi-part complex number problem

- 20 This question has two parts. Given $z_1 = 3 + 4i$ and $z_2 = 1 - 2i$. (a) Find $|z_1|$ (the modulus of z_1). (b) Find $z_1 \cdot z_2$ and its modulus, then verify that $|z_1 \cdot z_2| = |z_1| \cdot |z_2|$.



Scalar multiplication of matrices and vectors combined

- 21 Given vectors $\mathbf{u} = (4, -3)$ and $\mathbf{v} = (-2, 5)$, find $3\mathbf{u} - 2\mathbf{v}$.



Multiple choice — easy

- 22 Simplify $(10) + (4 - 2i)$.

- a. 14
- b. $-14 - 2i$
- c. $14 - 2i$
- d. $16 - 2i$

- 23 Simplify $(-10 + 6i) - (7 + 11i)$.

- a. $-17 - 5i$
- b. $-17 - 3i$
- c. $-15 - 5i$
- d. $17 - 5i$

24 Simplify $(3 + 9i) + (11 + 9i)$.

a. $-14 + 18i$

b. $14 + 20i$

c. $16 + 18i$

d. $14 + 18i$

25 Simplify $(2 - 9i) - (1 + 7i)$.

a. $3 - 16i$

b. $1 - 16i$

c. $-1 - 16i$

d. $1 - 14i$

26 Simplify $(6 - 10i) + (-9 + 8i)$.

a. $-1 - 2i$

b. $3 - 2i$

c. $-3 - 2i$

d. -3

27 Simplify $(-4 + 8i) - (-10 - 3i)$.

a. $8 + 11i$

b. $6 + 11i$

c. $-6 + 11i$

d. $6 + 13i$

28 Simplify $(-5 + 5i) - (9 - 10i)$.

a. $-14 + 17i$

b. $-14 + 15i$

c. $-12 + 15i$

d. $14 + 15i$

29 Simplify $(1 - 12i) + (8 + 5i)$.

a. $11 - 7i$

b. $-9 - 7i$

c. $9 - 7i$

d. $9 - 5i$

30 Simplify $(-9 - 12i) - (-12i)$.

a. -7

b. $-9 + 2i$

c. -9

d. 9

31 Simplify $(-1 + 3i) - (11 + 5i)$.

a. $-10 - 2i$

b. -12

c. $12 - 2i$

d. $-12 - 2i$

32 Simplify $(12 - 6i) + (-3 + 5i)$.

a. $9 - i$

b. $9 + i$

c. $11 - i$

d. $-9 - i$

33 Simplify $(-11 + 12i) + (8 + 11i)$.

a. $3 + 23i$

b. $-3 + 25i$

c. $-1 + 23i$

d. $-3 + 23i$

34 Simplify $(-8 + 10i) + (-9i)$.

a. $-8 + i$

b. $-8 + 3i$

c. $-6 + i$

d. $8 + i$

35 Simplify $(2 - i) + (-7 + 4i)$.

a. $5 + 3i$

b. $-5 + 5i$

c. $-3 + 3i$

d. $-5 + 3i$

36 Simplify $(-12i) + (3 - 6i)$.

a. $-3 - 18i$

b. $5 - 18i$

c. $3 - 16i$

d. $3 - 18i$

Multiple choice — medium

37 Simplify $(2 - 7i)(3 - 9i)$.

a. $-57 - 39i$

b. $-55 - 39i$

c. $69 - 39i$

d. $-57 + 3i$

38 Simplify $(-4 - 4i)(8 - 9i)$.

a. $-66 + 4i$

b. $4 + 4i$

c. $-68 + 4i$

d. $-68 + 68i$

39 Simplify $(-4 + 2i)(7 - 9i)$.

a. $-46 + 50i$

b. $-10 + 22i$

c. $-10 + 50i$

d. $-8 + 50i$

40 Simplify $(7 + i)(7 + 9i)$.

a. $42 + 70i$

b. $40 + 70i$

c. $40 + 56i$

d. $58 + 70i$

41 Simplify $(-3 + i)(-2 - 3i)$.

a. $11 + 7i$

b. $9 + 11i$

c. $9 + 7i$

d. $3 + 7i$

42 Simplify $(3 + 7i)(-5 - 6i)$.

a. $27 + 17i$

b. $27 - 53i$

c. $-57 - 53i$

d. $29 - 53i$

43 Simplify $(5i)(2 + 2i)$.

a. $10 + 10i$

b. $-10 - 10i$

c. $-10 + 10i$

d. $-8 + 10i$

44 Simplify $(-5 + 3i)(9 + 6i)$.

a. $-27 - 3i$

b. $-61 - 3i$

c. $-63 - 57i$

d. $-63 - 3i$

45 Simplify $(2 - 7i)(-7 - 5i)$.

a. $-47 + 39i$

b. $21 + 39i$

c. $-49 + 39i$

d. $-49 - 59i$

46 Simplify $(2 - 3i)(3 + 3i)$.

a. $15 + 15i$

b. $15 - 3i$

c. $17 - 3i$

d. $-3 - 3i$

47 Simplify $(8 - 2i)(-6 - 4i)$.

a. $-54 - 20i$

b. $-56 - 20i$

c. $-40 - 20i$

d. $-56 - 44i$

48 Simplify $(6 + 3i)(8 - 6i)$.

a. $66 - 60i$

b. $66 - 12i$

c. $30 - 12i$

d. $68 - 12i$

49 Simplify $(-3 + 6i)(4 - 2i)$.

a. $30i$

b. $-24 + 30i$

c. $2 + 30i$

d. $-18i$

50 Simplify $(-8 + 4i)(-2 - 9i)$.

a. $-20 + 64i$

b. $54 + 64i$

c. $52 + 64i$

d. $52 + 80i$

51 Simplify $(-6 + 8i)(-7 + 9i)$.

a. $-28 - 110i$

b. $-30 + 2i$

c. $-30 - 110i$

d. $114 - 110i$

Multiple choice — hard

- 52 Find the determinant of the matrix with rows $(4, 2)$ and $(3, 1)$.
- a. -4
 - b. 0
 - c. -2
 - d. 10
- 53 Find the determinant of the matrix with rows $(-4, 5)$ and $(7, -5)$.
- a. -13
 - b. -17
 - c. -3
 - d. -15
- 54 Find the determinant of the matrix with rows $(-1, -7)$ and $(-2, 4)$.
- a. -20
 - b. -18
 - c. 30
 - d. -16
- 55 Find the determinant of the matrix with rows $(-8, -9)$ and $(9, 8)$.
- a. 17
 - b. 15
 - c. 19
 - d. 0

56 Find the determinant of the matrix with rows $(-7, 6)$ and $(8, 6)$.

a. -92

b. -88

c. -90

d. -86

57 Find the magnitude of the vector $\mathbf{v} = (6, 8)$.

a. 10

b. 14

c. 9

d. 11

58 Find the magnitude of the vector $\mathbf{v} = (20, 21)$.

a. 29

b. 41

c. 30

d. 28

59 Find the magnitude of the vector $\mathbf{v} = (-3, 4)$.

a. 4

b. 7

c. 5

d. 6

- 60 Find the magnitude of the vector $\mathbf{v} = (20, -21)$.
- a. 28
 - b. 29
 - c. 41
 - d. 30
- 61 Find the magnitude of the vector $\mathbf{v} = (-7, -24)$.
- a. 26
 - b. 24
 - c. 25
 - d. 31
- 62 Simplify i^{16} , using the fact that powers of i cycle with period 4.
- a. i
 - b. $-i$
 - c. 1
 - d. -1
- 63 Simplify i^{36} , using the fact that powers of i cycle with period 4.
- a. 1
 - b. $-i$
 - c. i
 - d. -1

64 Simplify i^5 , using the fact that powers of i cycle with period 4.

a. -1

b. 1

c. $-i$

d. i

65 Simplify i^{40} , using the fact that powers of i cycle with period 4.

a. -1

b. i

c. $-i$

d. 1

66 Simplify i^{34} , using the fact that powers of i cycle with period 4.

a. 1

b. $-i$

c. i

d. -1